

Time: 1:30 hours

Marks: 30

[Answer all questions; Read the instruction below before answering the questions; Figures in right hand margin indicate full marks]

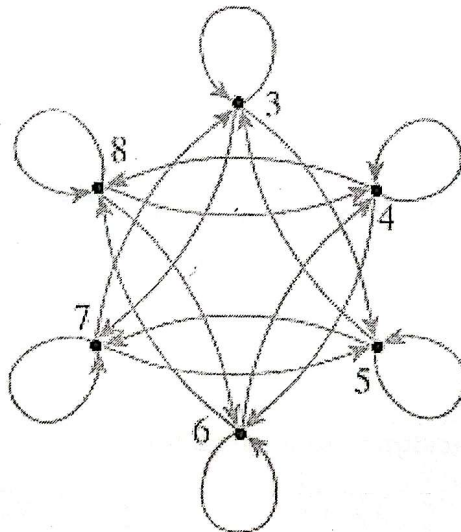
1. a) If the matrix $A = \begin{bmatrix} 4 & x+6 \\ 2x+3 & x+1 \end{bmatrix}$ is symmetric, find the value of x 2 CLO1 C2

② Or

Check whether it is singular matrix $\begin{bmatrix} 1 & 2 & d \\ 2 & 4 & f \\ 3 & 5 & a \end{bmatrix}$; Where a, d and f are the digits of your own ID number 2 CLO1 C2

- b) Find Inverse of the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ by making Linear Equations 3 CLO1 C2

- c) 3 CLO1 C2



Create an adjacency matrix for the above relation

- d) When do we use free variables when solving linear systems of equations? Give an example. 2 CLO1 C2

Or

Test whether $\lambda_1 = f$ and $\lambda_2 = e$ are Eigen values for $A = \begin{bmatrix} 5 & 3 \\ 2 & 3 \end{bmatrix}$; Where 2 CLO2 C2

e and f are the digits of your own ID number

②

2. a) Determine which of the following vectors are Eigen vectors for 5 CLO2 C2

$$A = \begin{pmatrix} 2 & e \\ -1 & -5 \end{pmatrix} \text{ showing your analysis procedure graphically}$$

(a) $\begin{pmatrix} a \\ f \end{pmatrix}$ (b) $\begin{pmatrix} b \\ e \end{pmatrix}$; Where a, b, e and f are the digits of your own ID number

(5)

- b) Determine whether the following vectors in R^3 are linearly dependent 5 CLO2 C2

or linearly independent $x_1 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$, $x_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$, $x_3 = \begin{bmatrix} 4 \\ 2 \\ -2 \end{bmatrix}$

Or

Verify Cayley Hamilton theorem for the matrix $\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$ and hence find A^{-1} .

5 CLO2 C2

(5)

3.

Diagonalize the matrix $A = \begin{bmatrix} 1 & 6 \\ 2 & 5 \end{bmatrix}$

10 CLO2 C2

Instruction:

If your ID no is C171023 then you can assign your digit as follows

C	1	7	1	0	2	3
	↓	↓	↓	↓	↓	↓
	2	4	3	2	3	2
	a	b	c	d	e	f

To solve the above problems, you need to use your own ID number where needed

International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
B. Sc. in CSE, Mid Term Examination, Spring-2025
Course Code: MATH-2307, Course Title: Mathematics-III

Time: 1:30 hours

Marks: 30

[Answer all questions; Read the instruction below before answering the questions; Figures in right hand margin indicate full marks]

1. a) If the matrix $A = \begin{bmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{bmatrix}$ is symmetric, find the value of x and hence present the matrix A 3 CLO1 C2
 Or
 Prove that $A = \begin{bmatrix} 2 & 2-3i & 3+5i \\ 2+3i & 3 & i \\ 3-5i & -i & 5 \end{bmatrix}$ is Hermitian 3 CLO1 C2
- b) Check whether it is singular matrix $\begin{bmatrix} 1 & 2 & d \\ 2 & 4 & e \\ 3 & 5 & f \end{bmatrix}$; Where d, e and f are the digits of your own ID number 2 CLO1 C2
- c) Name 5 methods of finding the inverse of a matrix 2 CLO1 C2
- d) Find the dependent row(s) from the following linear system of equations 1 CLO1 C2
 $2x + 3y + z = 1$
 $4x + 5y + 7z = 9$
 $x + 9y + z = 0$
 $4x + 6y + 2z = 2$
- e) Find the area of the parallelogram constructed by the vectors $\vec{u} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$ 2 CLO1 C2
 Or
 Test whether $\lambda_1 = f$ and $\lambda_2 = e$ are Eigen values for $A = \begin{bmatrix} 5 & 3 \\ 2 & 3 \end{bmatrix}$; 2 CLO2 C2
 Where e and f are the digits of your own ID number
2. a) Determine which of the following vectors are Eigen vectors for $A = \begin{pmatrix} 2 & f \\ -1 & -5 \end{pmatrix}$ showing your analysis procedure graphically 5 CLO2 C2
 $(a) \begin{pmatrix} a \\ e \end{pmatrix} (b) \begin{pmatrix} b \\ f \end{pmatrix}$; Where a, b, e and f are the digits of your own ID number

b) Determine whether the following vectors $\{(2, 1, 2), (0, 1, -1), (4, 3, 3)\}$ in R^3 are linearly dependent or linearly independent 5 CLO2 C2

Or

Verify Cayley Hamilton theorem for the matrix $A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$ and hence find A^{-1} . 5 CLO2 C2

3. Diagonalize the matrix $\begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$ 10 CLO2 C2

Instruction:

If your ID no is C171023 then you can assign your digit as follows

C	1	7	1	0	2	3
	↓	↓	↓	↓	↓	↓
	a	b	c	d	e	f

To solve the above problems, you need to use your own ID number where needed

International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
B. Sc. in CSE, Mid Term Examination, Spring-2024
Course Code: MATH-2307, Course Title: Mathematics-III

Time: 1:30 hours

Marks: 30

[Answer all questions; Read the instruction below before answering the questions; Figures in right hand margin indicate full marks]

1. a) If the matrix $A = \begin{bmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{bmatrix}$ is symmetric, find the value of x and hence present the matrix A 3 CLO1 C2

Or

- Prove that $A = \begin{bmatrix} 2 & 2-3i & 3+5i \\ 2+3i & 3 & i \\ 3-5i & -i & 5 \end{bmatrix}$ is Hermitian 3 CLO1 C2

- b) Check whether it is singular matrix $\begin{bmatrix} 1 & 2 & d \\ 2 & 4 & e \\ 3 & 5 & f \end{bmatrix}$; Where d, e and f are the digits of your own ID number 2 CLO1 C2

- c) Give an example of a Tri-diagonal matrix 1 CLO1 C2
d) Give one example of an Augmented Matrix 1 CLO1 C2
e) What is the relation between Eigen vector, Eigen value and given matrix 1 CLO1 C2

- f) Find the area of the parallelogram constructed by the vectors $\vec{u} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$ Or 2 CLO1 C2

- Test whether $\lambda_1 = f$ and $\lambda_2 = e$ are Eigen values for $A = \begin{bmatrix} 5 & 3 \\ 2 & 3 \end{bmatrix}$; Where e and f are the digits of your own ID number 2 CLO2 C2

2. a) Determine which of the following vectors are Eigen vectors for $A = \begin{pmatrix} 2 & f \\ -1 & -5 \end{pmatrix}$ showing your analysis procedure graphically 5 CLO2 C2

- (a) $\begin{pmatrix} a \\ e \end{pmatrix}$ (b) $\begin{pmatrix} b \\ f \end{pmatrix}$; Where a, b, e and f are the digits of your own ID number

b) Determine whether the following vectors in $\mathbb{R}^3$ are linearly dependent 5 CLO2 C2

or linearly independent $x_1 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, x_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, x_3 = \begin{bmatrix} 4 \\ 2 \\ -2 \end{bmatrix}$

Or

Verify Cayley Hamilton theorem for the matrix $A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$ and hence 5 CLO2 C2
find A^{-1} .

3.

Diagonalize the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

10 CLO2 C2

$$|A| = 8 + 3$$

Instruction:

If your ID no is C171023 then you can assign your digit as follows

C	1	7	1	0	2	3
	↓	↓	↓	↓	↓	↓
	a	b	c	d	e	f

To solve the above problem, you need to use your own ID number where needed

International Islamic University Chittagong
Department of Computer Science & Engineering
B.Sc. in CSE Mid Term Examination , Autumn -2023
Course Title: Mathematics-III Course Code: MATH-2307 (New)
Course Title: Mathematics-IV Course Code: MATH-2401 (Old)

Time: 1:30 hours

Marks: 30

[Please answer the following questions . Figures in the right margin indicates full marks]

1. a) Check whether it is singular matrix $A = \begin{bmatrix} 1 & e \\ 3 & f+1 \end{bmatrix}$; Where e and f are the digits of your

CLO2 2

own metric number

- b) Find the inverse of $A = \begin{bmatrix} 1 & e+1 \\ 3 & f+1 \end{bmatrix}$ making linear equations system; Where e and f are the

CLO2 3

digits of your own metric number

- c) Construct the 3x3 Matrix A having:

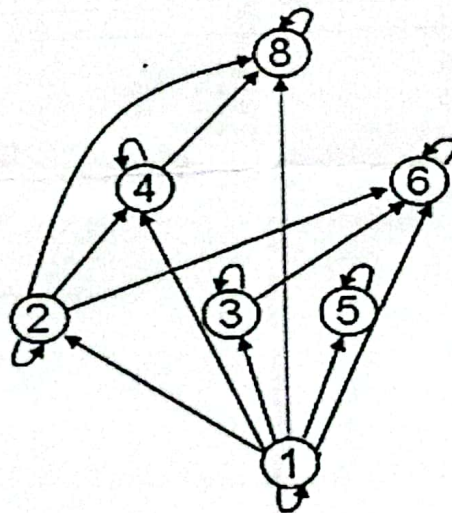
CLO1 3

$$A = (a_{ij}) = 4i + j ; \text{when } i > j$$

$$= d + 1 ; \text{when } i = j \text{ Where } d \text{ is the digit of your own metric number}$$

$$= 3i - j ; \text{when } i < j$$

Or



Create an adjacency matrix for the above figure

- d) Find the area of the parallelogram constructed by the vectors $\vec{u} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$

CLO2

Or

Test whether $\lambda_1 = f$ and $\lambda_2 = e$ are eigen values for $A = \begin{bmatrix} 5 & 3 \\ 2 & d \end{bmatrix}$; Where d, e and f are the digits of your own metric number

- e) What do you mean by a rank of a matrix?

CLO1

Or

Give an example of a tridiagonal Matrix

2. a)

Determine which of the following vectors are Eigen vectors for $A = \begin{pmatrix} 2 & d \\ 1 & -3 \end{pmatrix}$ showing your analysis procedure graphically

CLO2 5

(a) $\begin{pmatrix} 2 \\ e \end{pmatrix}$ (b) $\begin{pmatrix} f \\ 2 \end{pmatrix}$; Where d, e and f are the digits of your own metric number

b) Solve the system of linear equations

CLO2 5

$$x + y - z = 0,$$

$$2x + 4y - z = 0$$

$$3x + 2y + 2z = 0$$

Or

A manufacturer receives daily shipments of 70,000 springs and 45,000 pounds of stuffing for producing regular and support mattresses. Regular mattresses r requires 50 springs and 30 pounds of stuffing; support mattresses s requires 60 springs and 40 pounds of stuffing. How many mattresses of each type should be produced daily to use all available inventories?

3.

Diagonalize the matrix $A = \begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$

CLO2 10

Instructions:

If your matric no is C171023 then you can assign your digit as follows

C	1	7	1	0	2	3
	a	b	c	d	e	f

To solve the above problem, you need to use your own metric number where needed



International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
B. Sc. in CSE, Mid Term Examination (Special), Spring-2023
Course Code: MATH-2307, Course Title: Mathematics-III
Time: 1:30 hours Marks: 30
(Answer all questions. Figures in the right margin indicates full marks)
Section: 3AF

[N.B. Please answer the several parts of a question sequentially]

- | | | Marks | CLOs | DLs |
|-------|--|-------|------|-----|
| 1. a) | Define matrix. If $A = [a_{ij}]$ where $a_{ij} = \begin{cases} 0, & \text{when } i \neq j \\ 1, & \text{when } i = j \end{cases}$ then construct an 3×3 order matrix and Identify the type of matrix. Also test the matrix A is orthogonal or not. | 5 | CLO1 | C2 |
| b) | Define rank of a matrix and normal form of a matrix. Find the rank of the matrix, $A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$ after reducing it to the normal form. | 5 | CLO1 | C2 |
| Or. | What is the name of the following linear system of equations? Solve it and identify the type of solution
$x + y - z = 0$, $2x + 4y - z = 0$ and $3x + 2y + 2z = 0$ | | | |
| 2. a) | Define Eigen values and Eigen vectors. Check whether the vector $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$ are eigen vector for $A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$. Showing your analysis procedure graphically (using of graph paper is not mandatory) | 5 | CLO2 | C2 |
| Or. | Using matrix method, solve the following equations,
$x + y + z = 6$, $x - y + z = 2$ and $2x + y - z = 1$ | | | |
| b) | State Cayley-Hamilton. Verify the Cayley-Hamilton theorem hence find A^{-1} for $A = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$ | 5 | CLO2 | C2 |
| 3. | Define diagonal matrix. Find the eigenvalues and the corresponding eigenvectors of $A = \begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$ and hence diagonalize the matrix. | 10 | CLO2 | C2 |

International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
B. Sc. in CSE, Mid Term Examination, Autumn-2022
Course Code: MATH-2307, Course Title: Mathematics-III

Time: 1:30 hours

Marks: 30

[Please answer the questions serially. Figures in the right margin indicates full marks]

Marks CLO DL
5 CLO1 C2

1. a) If $A = [a_{ij}]$ where $a_{ij} = \begin{cases} 0, & \text{when } i \neq j \\ C, & \text{when } i = j \end{cases}$

Construct a 3×3 order matrix and Identify the type of matrix. Also test the matrix A is Orthogonal or not

Where C is the sum of the 1st digit & the last digit of your ID

- b) What is Augmented Matrix? Give one example

2 CLO1 C2

- c) Find the area of the parallelogram defined by the column matrices

2 CLO1 C2

$$u = \begin{bmatrix} -4 \\ 4 \end{bmatrix} \text{ and } v = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

- d) Name 4 methods of finding the inverse of a matrix

1 CLO1 C2

2. a) Check whether the vectors $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$ are Eigen vectors for $A = \begin{pmatrix} 1 & 6 \\ -4 & 7 \end{pmatrix}$ showing your analysis procedure graphically (using of graph paper is not mandatory)

3 CLO2 C2

- b) Prove that the set of vectors $\{(2, 1, 2), (0, 1, -1), (4, 3, 3)\}$ are linearly dependent.

3 CLO2 C2

- c) Prove that $A = \begin{bmatrix} 2 & 2-3i & 3+5i \\ 2+3i & 3 & i \\ 3-5i & -i & 5 \end{bmatrix}$ is Hermitian

2 CLO2 C2

- d) Determine whether $\lambda_1 = 5$ and $\lambda_2 = 4$ are eigen values for

2 CLO2 C2

$$A = \begin{bmatrix} 11 & 3 \\ -5 & -5 \end{bmatrix}$$

3. a) Using matrix method, solve the following equations, $x + y + z = 6$, $z - y + z = 2$ and $2x + y - z = 1$

5 CLO2 C2

- b) Verify the Cayley-Hamilton theorem and hence find A^{-1} for $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$

5 CLO2 C2

Or

3. Diagonalize the matrix $A = \begin{pmatrix} 1 & 4 \\ 4 & 1 \end{pmatrix}$

$$\begin{bmatrix} H - \lambda & 3 \\ -5 & -5 - \lambda \end{bmatrix}$$

10 CLO2 C2

$$\begin{bmatrix} 11 - \lambda & 3 \\ -5 & -5 - \lambda \end{bmatrix}$$

International Islamic University Chittagong
Department of Computer Science & Engineering
B.Sc. in CSE Online Mid Term Assessment , Spring -2022
Course Title: Mathematics-III Course Code: MATH-2307 (New)
Course Title: Mathematics-IV Course Code: MATH-2401 (Old)

Total Marks: 21

Section: 3AM+3BM

Time 2 hours 30 minutes for preparing assignment and 30 minutes for submission

Answer all the questions; ***Be careful to follow the instructions below*** for preparing and submitting the assignment; Figures in the right hand margin indicate full marks

1. a) Check whether it is singular matrix $\begin{bmatrix} 7 & 2 & f \\ 2 & 4 & e \\ 3 & 5 & 1 \end{bmatrix}$; Where d and e are the digits of your own metric number 2
- b) Find the inverse of $A = \begin{bmatrix} 1 & e \\ 3 & f+1 \end{bmatrix}$ Where d and f are the digits of your own metric number 3
- c) Construct the 4x4 Matrix A having: 2
 $A = (a_{ij}) = 4i + j$; when $i > j$
 $= 0$; when $i = j$
 $= 3i - j$; when $i < j$
2. Determine which of the following vectors are Eigen vectors for $A = \begin{pmatrix} 3 & d \\ -1 & -6 \end{pmatrix}$ showing your analysis procedure graphically 6
 $(a) \begin{pmatrix} 2 \\ e \end{pmatrix} (b) \begin{pmatrix} c \\ 2 \end{pmatrix} (c) \begin{pmatrix} e \\ f \end{pmatrix}$; Where a , b, c, d, e and f are the digits of your own metric number
 Which variable(s) is (are) free variables for the linear system of equations 1
 $7x + 5y + 9z = 6$
 $w = 1$
3. a) Test whether $\lambda_1 = f$ and $\lambda_2 = e$ are eigen values for $A = \begin{bmatrix} 5 & 3 \\ 2 & 3 \end{bmatrix}$; Where d and f are the digits of your own metric number 2
- b) Solve the system of linear equations 5
 $(e + 2)x + 9y - z = 27$
 $x - 8y + 16z = 10$ Where a and f are the digits of your own
 $ax + y + 15z = 37$
 metric number

Instructions:

If your matric no is C171023 then you can assign your digit as follows

C	1	7	1	0	2	3
	↓	↓	↓	↓	↓	↓
	a	b	c	d	e	f

To solve the above problem, you need to use your own metric number where needed